Lecture 14 (*March 19*)

Reading: Section 4.4
Viewing: Lecture 17

Question: when are two planes V and W orthogonal? Answer: when any two vectors $\boldsymbol{v} \in V$ and $\boldsymbol{w} \in W$ are orthogonal. Of course, it is hard to check this property for any vectors, but it suffices to check it for all vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ and $\boldsymbol{w}_1, \dots, \boldsymbol{w}_l$ which form bases of V and W, respectively:

$$V \perp W \quad \Leftrightarrow \quad \boldsymbol{v}_i^T \boldsymbol{w}_j = 0 \text{ for all } i \in \{1, \dots, k\},\ j \in \{1, \dots, l\} \tag{113}$$

Since there are k basis vectors of V and l basis vectors of W, there are $k \cdot l$ conditions to check above. You can replace this with a single condition, if you put the $\boldsymbol{v}$'s together in a matrix A and the $\boldsymbol{w}$'s together in a matrix B:

$$A = \left[\ \boldsymbol{v}_1 \mid \dots \mid \boldsymbol{v}_k\ \right] \qquad\qquad B = \left[\ \boldsymbol{w}_1 \mid \dots \mid \boldsymbol{w}_l\ \right]$$

Then $V = C(A)$ and $W = C(B)$, and condition (113) reads:

$$V \perp W \quad \Leftrightarrow \quad A^T B = 0 \tag{114}$$

The following is a related notion, which applies to bases of a single vector space V.

Definition 14. *A collection of non-zero vectors $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$ are called **orthogonal** if:*

$$\boldsymbol{q}_i \perp \boldsymbol{q}_j \quad \textit{or, equivalenty} \quad \boldsymbol{q}_i^T \boldsymbol{q}_j = 0 \quad \textit{for all } 1 \leq i \neq j \leq n \tag{115}$$

*It is called **orthonormal** if the vectors are both orthogonal and have length 1:*

$$||\boldsymbol{q}_i|| = 1 \qquad \textit{or, equivalenty} \qquad \boldsymbol{q}_i^T \boldsymbol{q}_i = 1 \quad \textit{for all } 1 \leq i \leq n \tag{116}$$

Orthogonal vectors are always linearly independent (so a basis of the vector space they span).

The standard basis $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$ of $\mathbb{R}^n$ (where $\boldsymbol{e}_i = (0, \dots, 0, 1, 0, \dots, 0)$ with the 1 on the i–th spot) is orthonormal. In fact, the notion of orthonormality seeks to generalize this behavior to other bases.

There are n^2 conditions (115)–(116), so there are n^2 properties to check in order to show that a collection of n vectors are orthogonal. As before, this can be packaged into a single matrix equality, if we consider the $m \times n$ matrix (assume $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n \in \mathbb{R}^m$):

$$Q = \left[\ \boldsymbol{q}_1 \mid \dots \mid \boldsymbol{q}_n\ \right]$$

Then the orthogonality property (115) precisely states that:

$$Q^T Q = \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} \tag{117}$$

where $d_i = ||\boldsymbol{q}_i||^2$. Therefore, the orthonormality properties (115)–(116) precisely state that:

$$\boxed{Q^T Q = I_n} \tag{118}$$

In the particular case when $m = n$, we have:

Definition 15. *A square matrix Q is called* ***orthogonal*** *if* (118) *holds, i.e. if* $Q^T = Q^{-1}$.

Very important: Definition 15 (as well as the term "orthogonal matrix") only applies to square matrices. This is because while we can state property (118) for a non-square matrix, it does not make sense to take the inverse of a non-square matrix.

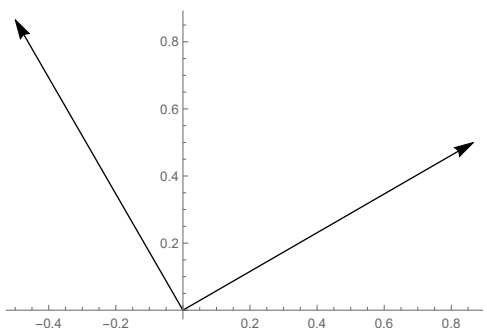
Examples of orthogonal matrices include all permutation matrices:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

and the rotation matrices:

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \tag{119}$$

The columns of the matrix Q of (119) are the usual unit vectors in $\mathbb{R}^2$, rotated by an angle θ:



and they are orthonormal because $(\sin\theta)^2 + (\cos\theta)^2 = 1$. Both examples above share a feature, which is actually common of all orthogonal matrices: they preserve dot products, lengths and perpendicularity of vectors:

$$\text{if } Q \text{ is an orthogonal matrix, then } \boxed{(Q\boldsymbol{v})^T(Q\boldsymbol{w}) = \boldsymbol{v}^T\boldsymbol{w}} \tag{120}$$

for all vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ of the same size. The reason for this is the chain of equalities:

$$(Q\boldsymbol{v})^T(Q\boldsymbol{w}) = \boldsymbol{v}^T Q^T Q\boldsymbol{w} = \boldsymbol{v}^T I\boldsymbol{w} = \boldsymbol{v}^T\boldsymbol{w}$$

Therefore, as a consequence of (120), we have:

$$\text{if } \boldsymbol{v} \perp \boldsymbol{w} \quad \text{then} \quad Q\boldsymbol{v} \perp Q\boldsymbol{w} \tag{121}$$

and:

$$||Q\boldsymbol{v}|| = ||\boldsymbol{v}|| \tag{122}$$

for all vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ of the same size.

The discussion above was for square matrices, but now let's consider general ones. More precisely, let's return to the projection formula (103), which is generally cumbersome because we need to

invert the matrix $A^T A$. In the particular case when $A = Q$ has orthonormal columns (i.e. we have an orthonormal basis of the vector space we are projecting upon), the projection formula is nicer:

$$P_{C(Q)} = QQ^T \tag{123}$$

because as we have seen in (118):

$$Q \text{ has orthonormal columns} \quad \Leftrightarrow \quad Q^T Q = I$$

Similarly with the discussion above, in the case when $A = Q$ has orthonormal columns, the solution to the least squares problem (109) simplifies to:

$$\boldsymbol{v} = Q^T \boldsymbol{b} \tag{124}$$

To summarize: if you're projecting onto a subspace with general basis vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$, then you need to apply the more complicated formulas (103), (109) for $A = \big[\boldsymbol{v}_1 | \ldots | \boldsymbol{v}_n\big]$. But if you know that your subspace has a basis consisting of orthonormal vectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$, then you may apply the simpler formulas (123), (124) for $Q = \big[\boldsymbol{q}_1 | \ldots | \boldsymbol{q}_n\big]$. It pays to have orthonormal bases of subspaces.

Let's do an example of computing the projection matrix using formula (123) instead of the original formula (103). Suppose you wanted a formula for the projection matrix onto the plane:

$$V = \Big\{ x + y + z = 0 \Big\} \subset \mathbb{R}^3 \tag{125}$$

The "old" way of doing this was to pick a basis of V (i.e. any two vectors in V which are not multiples of each other), form the 3×2 matrix A whose columns are the given vectors, and apply formula (103). The "new" way is to start not from an arbitrary basis, but from an orthonormal basis of V. We'll learn how to construct such bases systematically in the next lecture, but the first step is to pick two orthogonal (i.e. perpendicular) vectors in V. For example, the vector:

$$\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

lies in V, so it can be the first vector in our basis. Now we need to pick a second vector $\boldsymbol{v}_2$ which is at the same time orthogonal to $\boldsymbol{v}_1$ (i.e. the first two coordinates of $\boldsymbol{v}_2$ must be equal to each other) and also lie in V (i.e. the coordinates of $\boldsymbol{v}_2$ must sum up to 0). A natural choice then is:

$$\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

To turn an orthogonal basis into an orthonormal basis, we need to rescale the vectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ so that they have length 1. This is achieved by dividing $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ by their lengths, which are:

$$||\boldsymbol{v}_1|| = \sqrt{2} \qquad \text{and} \qquad ||\boldsymbol{v}_2|| = \sqrt{6}$$

Therefore, we conclude that the vectors:

$$\boldsymbol{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{q}_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \tag{126}$$

form an orthonormal basis of V. Then we form a matrix whose columns are the orthonormal basis:

$$Q = \left[\begin{array}{c|c} \boldsymbol{q}_1 & \boldsymbol{q}_2 \end{array}\right] = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$$

and formula (123) tells us that the projection matrix we're looking for takes the form:

$$P_V = QQ^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{-2}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{bmatrix}$$